

University Denomination Directory

Requirements Document — v2 (working title; final product name pending)

1. Problem

- Students, especially freshers, struggle to locate the religious denomination they are looking for on campus.
- Denominations currently recruit by walking around campus and manually collecting names and phone numbers from students, then adding them to a WhatsApp group.
- Meeting information (location, day, time) is rarely available anywhere centrally — it spreads by word of mouth.

2. Solution Summary

A decentralized web platform where students register directly into their denomination's specific campus chapter and are instantly routed into the correct WhatsApp group — replacing manual, on-foot recruitment. Each chapter is run by its own Chapter Head, who manages that chapter's roster and details. A System Admin manages the overall structure (denominations, chapters, head assignments) without day-to-day access to member data.

3. Roles & Permissions

Access is scoped at the chapter level, not the denomination level. A Chapter Head only ever sees data for their own chapter, even if another chapter shares the same denomination name.

Role	Can do	Cannot do
Student	Browse chapters (own campus), register into one chapter, view/edit own profile, leave chapter (self-service)	View other students' data; belong to more than one chapter at a time
Chapter Head	View & manage their chapter's roster; edit their chapter's profile (meeting info, WhatsApp link); flag/remove a member	View another chapter's roster, even within the same denomination
System Admin	Create denominations and chapters; assign/remove Chapter Heads; view aggregate/structural data	View individual student rosters or contact details by default

4. Data Model

Entities and key fields. See accompanying diagram for relationships (Denomination → Chapter → Head / Students).

4.1 Denomination

Field	Type / Notes
id	Primary key
name	e.g. "Pentecost Fellowship"
description	Optional

4.2 Chapter

Field	Type / Notes
id	Primary key
denomination_id	Foreign key → Denomination
campus	Main Campus or Essikado
status	draft / coming soon / active
meeting_day	Set by Chapter Head
meeting_time	Set by Chapter Head

location	Set by Chapter Head
whatsapp_link	Invite link generated by WhatsApp itself when the Head manually creates the group; the Head pastes it in — the platform does not generate or touch WhatsApp groups

4.3 Head (Chapter Admin)

Field	Type / Notes
id	Primary key
chapter_id	Foreign key → Chapter (one head per chapter)
name	
phone / email	Used for password-based login

4.4 Student

Field	Type / Notes
email	Primary key / unique ID — also the OTP login identifier
phone_number	Contact field only — no longer the identity key or exclusivity check
name	
program	
level	Used for graduation-based data retention
hall / hostel	
current_chapter_id	Foreign key → Chapter; null if unregistered. Enforces single-chapter exclusivity

5. Authentication Model

- Single login entry point for all users. Role and chapter scoping are resolved server-side after authentication — there is no separate login screen per role.
- Students: passwordless. Email + OTP verifies identity — OTP is delivered via email (Google SMTP) for v1, with a planned upgrade to SMS OTP later. A long-lived session token keeps them signed in so they rarely need to re-verify.
- Email is now the student's unique identifier and login credential. Phone number is still collected at registration as a contact field, but no longer determines identity or the single-chapter exclusivity check.
- Chapter Heads and System Admin: password-based accounts. Justification — a compromised admin account exposes an entire chapter's roster, so the smaller, more tech-comfortable admin population absorbs slightly more login friction in exchange for stronger protection of aggregated student data.

6. User Flows

6.1 Student Flow

- Sign up with email address, verify via OTP sent to that email (Google SMTP for v1).
- Select campus (Main Campus or Essikado) — this filters which chapters are shown.
- Browse / search chapters. Chapters still being set up show as "coming soon" (no meeting info or WhatsApp link yet, but visible).
- Select a chapter, fill registration form: Name, Phone Number, Campus, Program, Hostel/Hall, Level.
- System checks the student's email against existing registrations. If already registered elsewhere, registration is blocked with an error naming the current chapter.
- On success, the WhatsApp invite link is returned instantly — no approval wait.
- Student taps through and joins the WhatsApp group.
- Later: student can view/edit their own profile, or leave their chapter (self-service), which frees their email to register elsewhere.

6.2 Chapter Head Flow

- Receives account (password-based) from System Admin, tied to one specific chapter.
- Completes the chapter's operational profile: meeting day/time, location, description, and WhatsApp invite link. This flips the chapter's status from draft/coming-soon to active.
- Ongoing: views their chapter's member roster; flags or removes a member (removal from the roster is separate from removal from WhatsApp itself, which currently remains a manual step the Head performs directly in WhatsApp).
- Edits chapter profile details as meeting logistics change.

6.3 System Admin Flow

- The first System Admin account is seeded directly at system setup — it cannot be created through normal signup, since there is no higher authority to approve it.
- A fellowship leader contacts the System Admin directly, outside the app (no in-app request form for v1).
- System Admin creates the Denomination (name, description).
- System Admin creates a Chapter shell under that denomination for a specific campus. Status starts as draft / coming soon — visible to students as a placeholder, but not yet joinable.
- System Admin creates the Chapter Head's account and assigns it to that chapter.
- Chapter Head logs in and completes the chapter profile (see 6.2), which activates the chapter.
- Ongoing: System Admin manages structure only — denominations, chapters, head assignments — without default access to individual student rosters.

7. Chapter Lifecycle

A chapter moves through three states:

- Draft — created by System Admin, no head assigned yet. Not shown to students.
- Coming soon — head assigned, but meeting info and/or WhatsApp link not yet complete. Shown to students as a placeholder (name, campus, "coming soon") but not joinable.
- Active — meeting info and WhatsApp link complete. Fully visible, searchable, and joinable in the student directory.

8. Data Privacy & Retention

- Access to student data is scoped to the owning Chapter Head only; the System Admin does not have blanket access to student rosters by default.
- Email is the platform-wide unique identifier for students and enforces single-chapter membership. Phone number is retained as contact data only.
- Students recorded at level 400 are planned to be auto-removed from the system roughly one year later (graduation-based retention), which also frees their phone number for reuse.
- No formal identity verification exists yet for Chapter Heads — the System Admin adds them based on direct, informal contact (e.g. an outgoing head vouching for their successor).
- This platform collects personal data (names, phone numbers, program, hall) at scale, which falls under Ghana's Data Protection Act (Act 843) — registration with the Data Protection Commission and basic safeguards (purpose limitation, consent, security) should be considered before scaling past a pilot.

9. Open Items — Not Yet Decided

Flagged during discussion but intentionally left open for a later decision:

- Whether a student's long-lived session should be force-expired when they leave a chapter, or simply persist until natural expiry.

- Google SMTP's sending limit (roughly 500 emails/day on a standard account) as a ceiling to watch for — a signal for when to actually make the planned SMS upgrade, not just a nice-to-have.
- Password recovery flow for Chapter Heads / System Admin (a minimal reset-via-email flow is the likely default).
- Whether to track membership history (e.g. "previously in the Essikado chapter") rather than only the current chapter, which the present model does not retain.
- Whether a "coming soon" chapter should let students register interest (capturing a phone number ahead of the chapter going active) so the new Head has a warm list to notify at launch.